Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories, customer segments, and subscription behavior.

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Dataset at a Glance

3,900

Transactions

Rows in dataset

18

Features

Columns tracked

37

Missing Values

Review Rating column

$50.76

Avg Purchase

Mean spend (USD)

Key features: demographics, purchase details, shopping behavior, subscription status.

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Data Preparation in Python

Cleaning

Imputed missing Review Ratings using category medians; standardized columns to snake_case.

Feature Engineering

Created **age_group** bins and **purchase_frequency_days**; dropped redundant promo_code_used.

Database Load

Connected to PostgreSQL; loaded cleaned DataFrame for SQL analysis.

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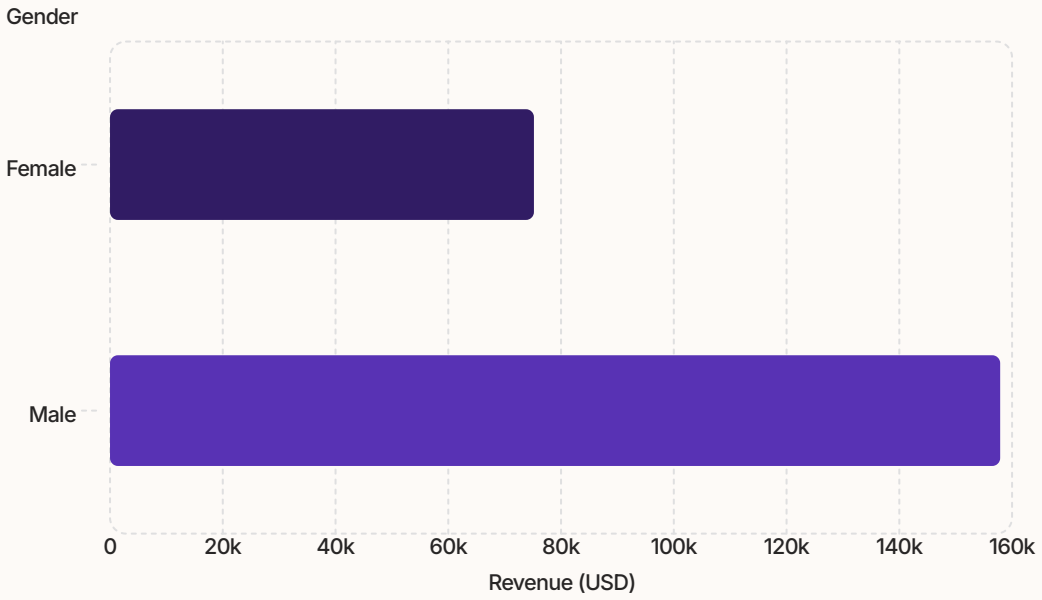
Revenue by Gender



Key Insight

Male customers generated **2.1× more revenue** than female customers — $157,890 vs. $75,191.

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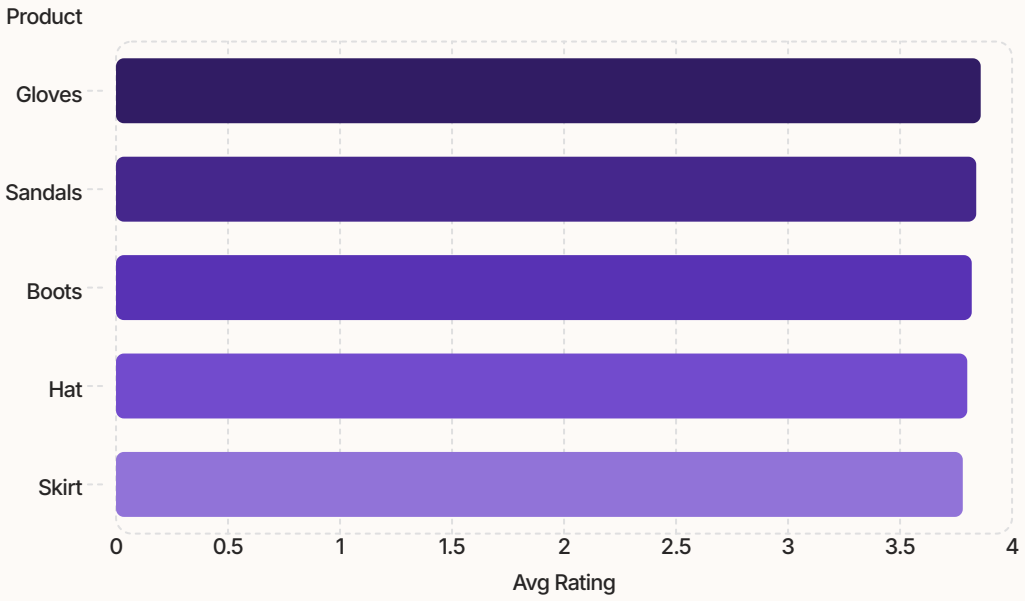
Top Products by Rating



Overall Avg Rating

Dataset average: **3.75 / 5.0**

Top-rated items cluster tightly between 3.78–3.86.

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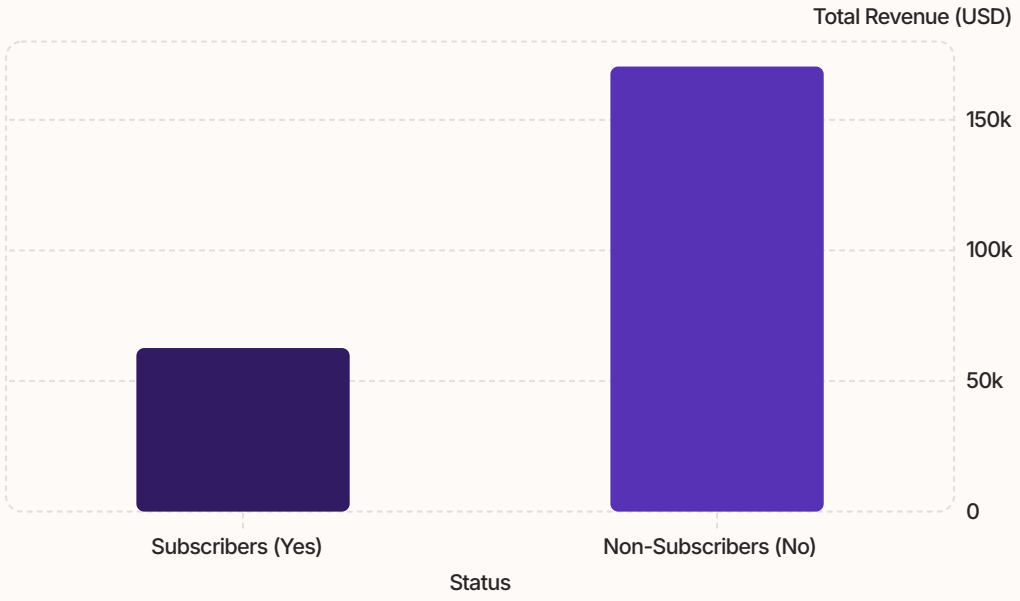
Subscribers vs. Non-Subscribers



Subscription Snapshot

- **27%** subscribers — 1,053 customers
- **73%** non-subscribers — 2,847 customers
- Avg spend nearly identical: ~$59.50

Non-subscribers drive **73% of total revenue** despite similar per-customer spend.

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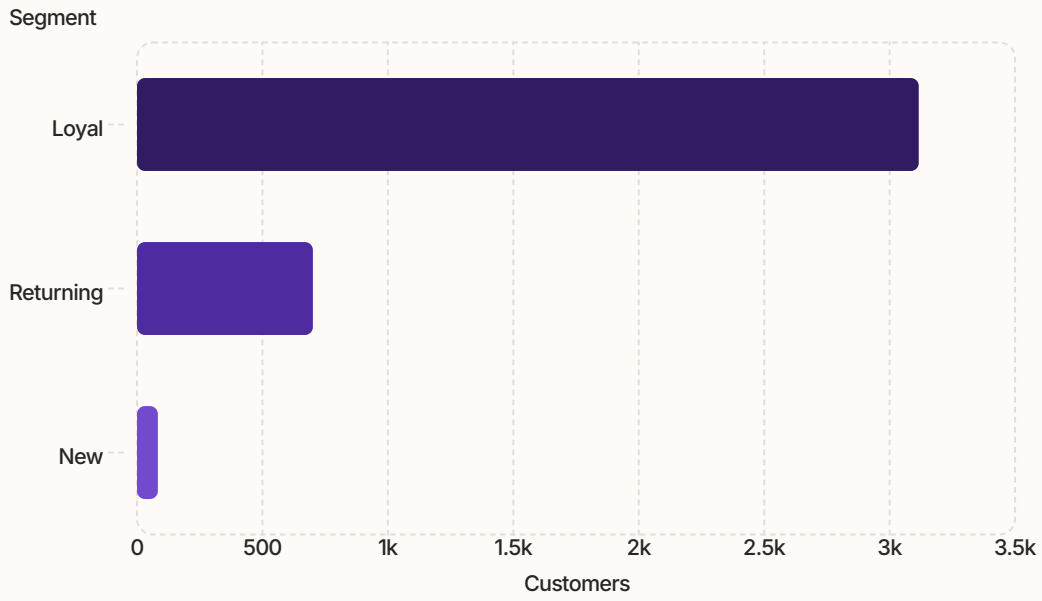
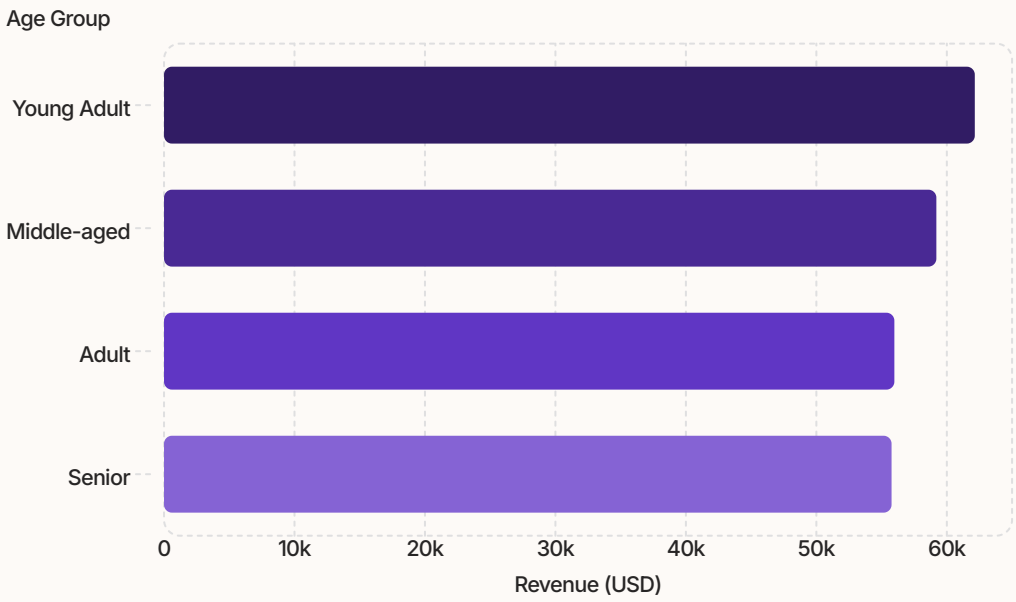
Customer Segmentation & Revenue by Age



Segment Breakdown

80% Loyal · 18% Returning · 2% New

Revenue by Age Group



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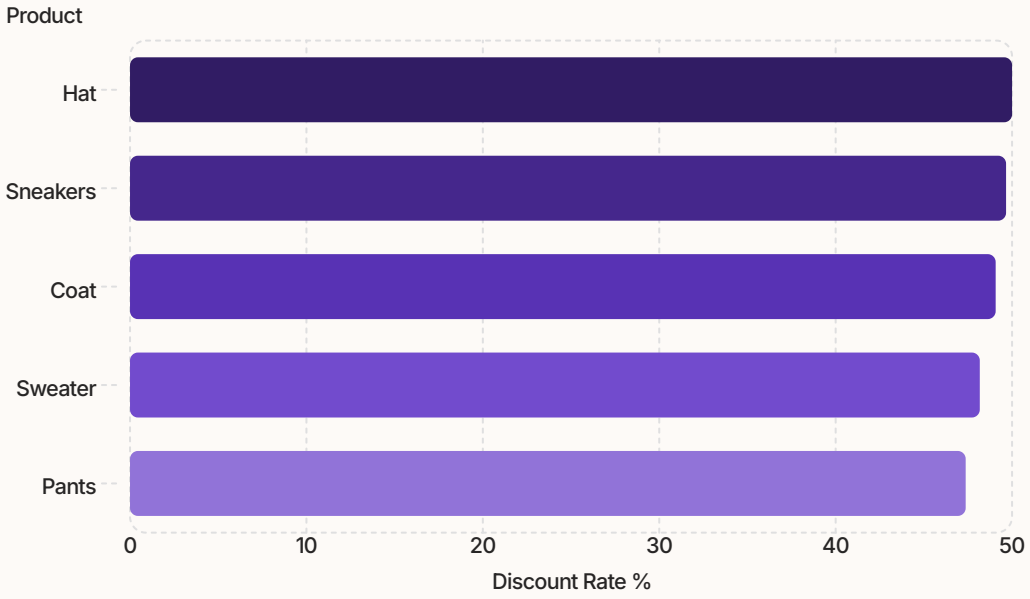
Discount Behavior & Shipping

Top Discount-Dependent Products



Shipping Comparison

Standard

Avg $58.46

Express

Avg $60.48

Express buyers spend slightly more on average.

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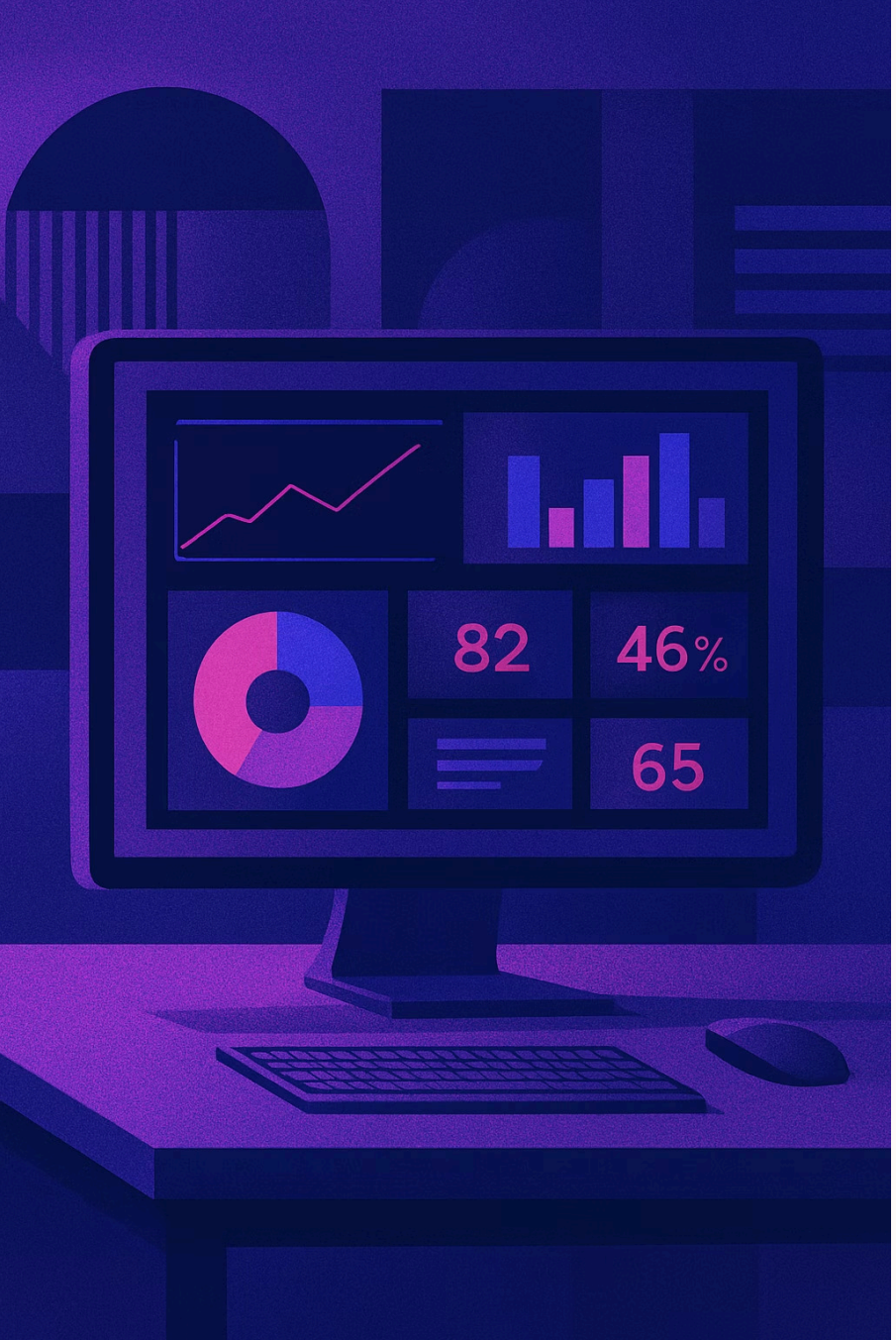

Power BI Dashboard

3.9K Customers

Total tracked

$59.76 Avg

Purchase amount

3.75 Avg

Review rating

Interactive filters: Subscription Status, Gender, Category, Shipping Type. Visualizations include revenue by category, age group, and subscription split.

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Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits.

Loyalty Programs

Reward repeat buyers toward "Loyal" segment.

Review Discount Policy

Balance sales lift with margin control.

Targeted Marketing

Focus on high-revenue age groups and express-shipping users.

Product Positioning

Highlight top-rated and best-selling products in campaigns.

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